

ACC Two-Channel Torque Assembly

Physical Conditions — Smoothstep Gate Inputs

Proximity $|\Delta x|$ · Quietness $\text{EMA}(|v|)$ · Stability θ ,
 θ · Contact Fz^L , Fz^R · Terrain h^L_{gnd} , h^R_{gnd}

Wheel Torque τ_w

τ_{balance}
Sagittal Balance Core
(pitch PD + vel. damp + pos-P
+ lateral + yaw)
always active

+

$g_{\text{anchor}} \cdot \tau_{\text{anchor}}$
Proximity-Gated Anchor
(integral + damping boost)
gated by g_{prox} , g_{env}

+

$g_{\text{flight}} \cdot \tau_{\text{flight}}$
Contact-Loss Recovery
(reaction-wheel attitude PD)
gated by g_{flight}



$$\tau_w = \tau_{\text{bal}} + g_{\text{anc}} \cdot \tau_{\text{anc}} + g_{\text{flt}} \cdot \tau_{\text{flt}}$$

Leg-Joint Torque τ_g

$\tau_{\text{posture}}(h_{\text{cmd}})$
Posture & Yaw Stability
(Jacobian PD + hip-yaw
divergence + homing)
always active, height-scheduled

+

$g_{\text{terrain}} \cdot \Delta \tau_{\text{posture}}$
Per-Leg Ground Adaptation
(split height commands
+ leveling integrator)
gated by g_{terrain}



$$\tau_g = \tau_{\text{post}} + g_{\text{terr}} \cdot \Delta \tau_{\text{post}}$$

Smoothstep Gate Definitions

$$g_{\text{prox}} = 1 - \text{ss}(|\Delta x|, 0.05, 0.15) \quad g_{\text{env}} = 1 - \text{ss}(\text{EMA}(|v|), 0.18, 0.30)$$

$$\text{Asymmetric EMA: } \alpha_{\text{attack}} = 0.35 \ (\tau \approx 30\text{ms}), \\ \alpha_{\text{release}} = 0.007 \ (\tau \approx 1.5\text{s}) \quad k_p = 50 - 15 \cdot (1 - g_{\text{env}} \cdot g_{\text{prox}})$$

$$g_{\text{flight}}: Fz < 0.5\text{mg for } \geq 2 \text{ steps, release } \geq 5 \text{ steps} + 150\text{ms ramp} \quad g_{\text{terrain}}: |\Delta h| > 2\text{cm, unloaded freeze}$$

Both channels assembled independently $\rightarrow$ final 10-DoF torque command

Wheel torque	Channel output	Gate definition
Leg-joint torque	Physical condition	Scheduled gain